AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

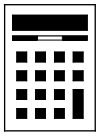
# Functional Skills Level 2
# MATHEMATICS

## Paper 2 Calculator

Thursday 3 November 2022 Afternoon Time allowed: 1 hour 30 minutes

**Materials**

For this paper you must have:
- a calculator
- mathematical instruments.



| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–8 | |
| 9 | |
| 10 | |
| 11 | |
| 12 | |
| **TOTAL** | |

**Instructions**
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

**Information**
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a $\pi$ button, take the value of $\pi$ to be 3.142

**Advice**

In all calculations, show clearly how you work out your answer.

*Do not write outside the box*

## Section A

Answer **all** questions in the spaces provided.

**1** Circle the integer.

**[1 mark]**

0.5 $\frac{1}{8}$ 7 $-10.2$

**2** Write 9 507 211 in words.

**[1 mark]**

Answer _____

**3** Work out 3 years to 9 months as a ratio.

Give your answer in its simplest form.

**[2 marks]**

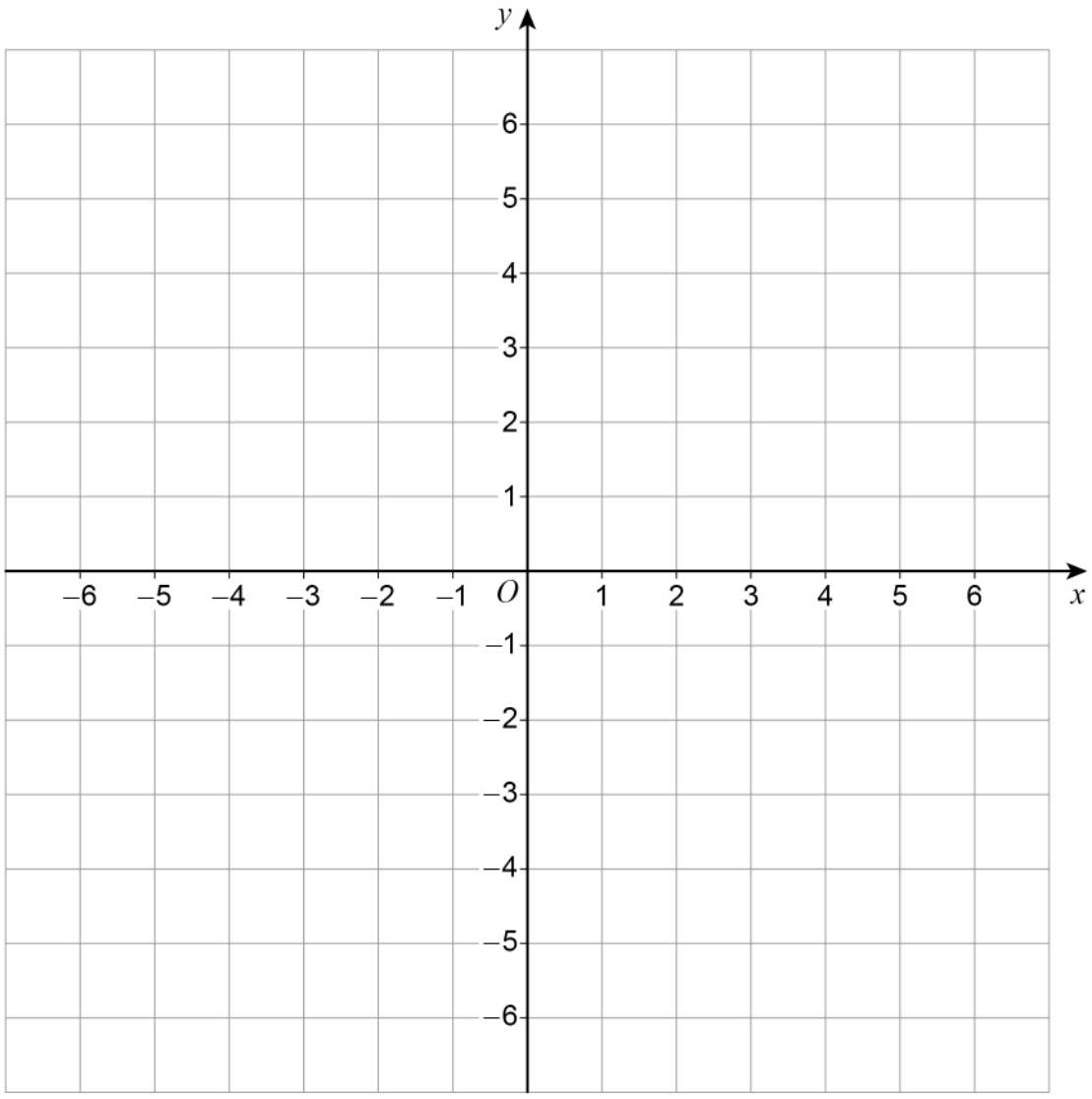
Answer _____ : _____

*Do not write outside the box*

**4** On the grid, plot and label the points $X$, $Y$ and $Z$.

**[2 marks]**

$X = (3, 5)$ $Y = (5, -3)$ $Z = (-3, -5)$

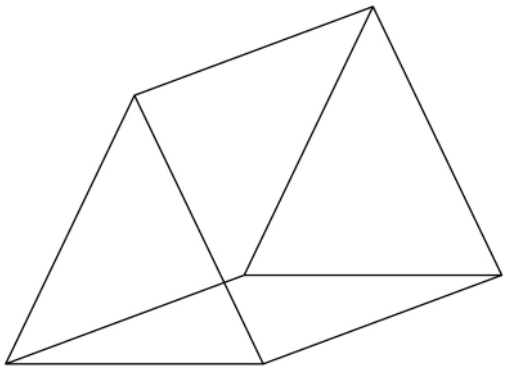



**Turn over for the next question**

*Do not write outside the box*

**5** Write the mathematical name of this solid shape.

**[1 mark]**



Answer _____

**6** Calculate $2\frac{1}{5} + 1\frac{3}{4}$

**[1 mark]**

_____

_____

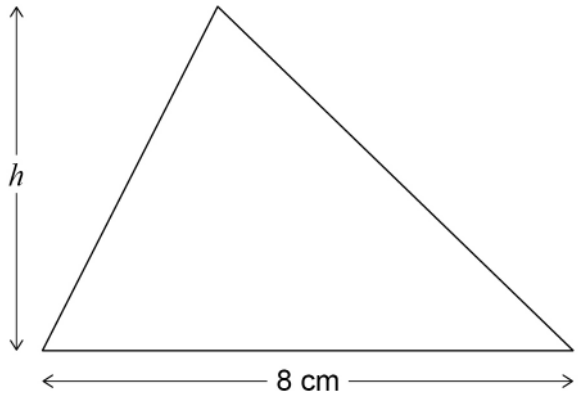
Answer _____

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**7** A triangle has an area of 20 $cm^2$

The base of the triangle is 8 cm

Not drawn accurately




Work out the perpendicular height, $h$, of the triangle.

**[2 marks]**

Answer _____ cm

**8** Calculate $2(7 + 3k)$ when $k = -1.8$

**[2 marks]**

Answer _____

**12**

**Turn over ►**

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## Section B

Answer **all** questions in the spaces provided.

**9** **Lorry driving**

Asha is a lorry driver.

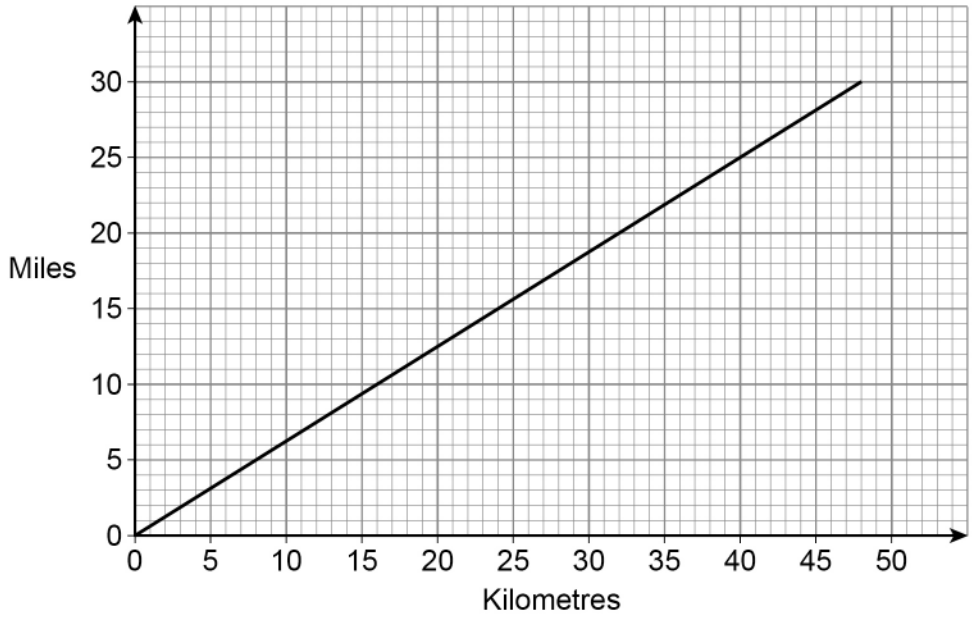
**9 (a)** Asha drives from Southampton to Leeds.

The journey

is 380 kilometres

takes 5 hours and 30 minutes.

The graph can be used to convert between miles and kilometres.




Asha works out that his average speed is **over** 40 mph

Is he correct?

You **must** show your working.

**[4 marks]**

*Do not write outside the box*

**9 (b)** The amount Asha is paid each week is calculated using the formula

$$P = 0.73(0.14d + 65n)$$

where

$P$ = pay in pounds

$d$ = distance driven in kilometres that week

$n$ = number of days worked that week

Last week Asha worked for 5 days.

His pay for last week's work was £605.17

How many kilometres did Asha drive last week?

**[4 marks]**

Answer _____ kilometres

8

**Turn over for the next question**

*Do not write outside the box*

**10 Fundraising**

Carol is fundraising for a sports club.

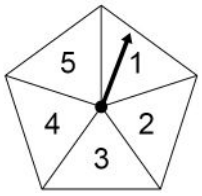
**10 (a)** Carol designs a game.

The game uses

a bag containing a red ball, a blue ball and a yellow ball

a fair, 5-sided spinner.




The player

picks a ball at random from the bag

and

spins the spinner.

The player wins if they pick the **red** ball and the spinner lands on an **even** number.

Carol says,

"The chance of winning is **more than** 10%"

Is she correct?

You **must** show your working.

**[4 marks]**

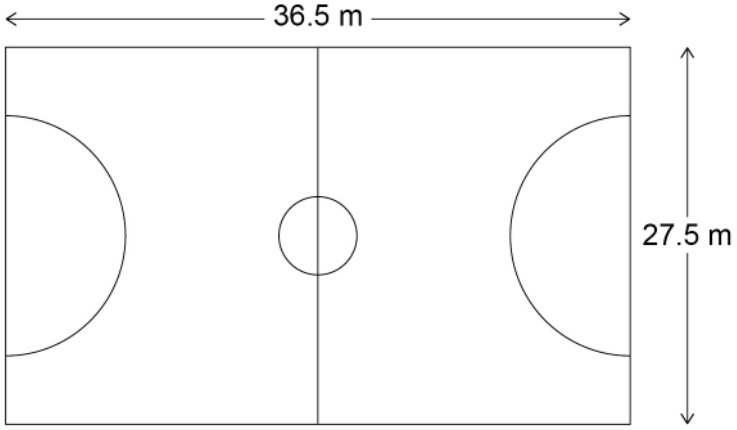
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**10 (b)** The club wants to use some of the money to paint the lines on a mini football pitch.

The lines to be painted are

- the four sides of a rectangle measuring 27.5 m by 36.5 m
- a halfway line measuring 27.5 m
- a centre circle with a radius of 1.5 m
- two semicircles, each with a radius of 8 m

Not drawn accurately




It costs £3.25 per metre to paint the lines.

In total, how much will it cost to paint all the lines?

**[6 marks]**

Answer £ _____

**Turn over ►**

*Do not write outside the box*

**10 (c)** After painting the lines the club has £8225

They invest $\frac{2}{7}$ of this money in a bank account for 4 years.

The account pays compound interest at 3% per year.

Is the investment worth **more than** £2700 at the end of the **4** years?

You **must** show your working.

**[4 marks]**

**14**

*Do not write outside the box*

**11** **Ice cream**

Suzi has an ice cream van.

**11 (a)** Suzi buys tubs of ice cream and sells scoops of ice cream.

Each scoop is in the shape of a sphere with radius 2.8 cm

$$\text{volume of sphere} = \frac{4}{3}\pi r^3$$

$r$ = radius of sphere

Suzi buys 5-litre tubs.

1 litre = 1000 $cm^3$

Suzi wants to buy enough tubs to sell at least 200 scoops.

Work out how many tubs Suzi should buy.

You **must** show your working.

**[5 marks]**

Answer _____

**Question 11 continues on the next page**

*Do not write outside the box*

**11 (b)** Suzi uses a 15% discount voucher when she buys the tubs of ice cream.

She pays £76.50 after the discount.

Suzi says,

"I save **less than** £14 by using the discount voucher."

Show working to support this statement.

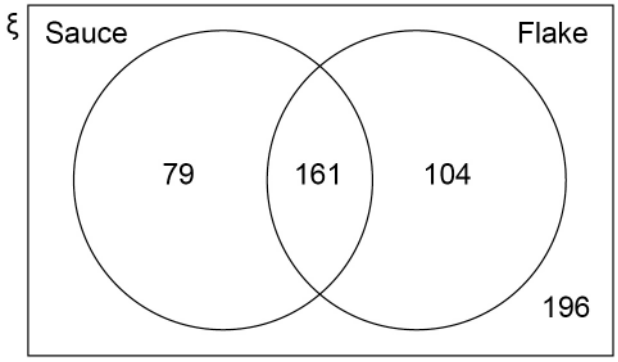
**[3 marks]**

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**11 (c)** Suzi sells two ice cream toppings, sauce and flake.

She hopes that the probability that a customer, picked at random, buys **at least one** topping will be more than $\frac{7}{10}$

The Venn diagram shows what toppings the customers buy over one weekend.




Over this weekend, does Suzi achieve the probability she hopes to get?

You **must** show your working.

**[3 marks]**

**11**

**Turn over for the next question**

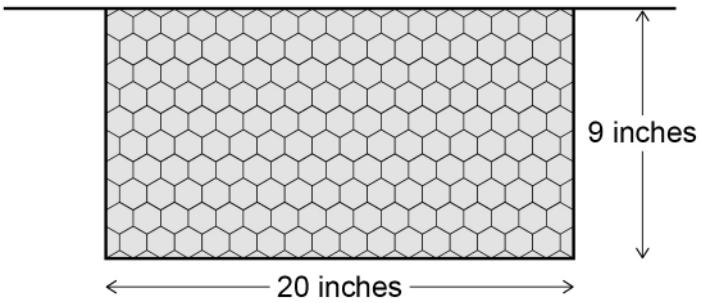
*Do not write outside the box*

**12** **Bees**

Mary keeps bees and sells the honey they produce.

**12 (a)** The bees live in a beehive.

Mary's beehive holds rectangular frames full of honeycomb.




Not drawn accurately

Each frame measures 9 inches by 20 inches.

The beehive holds 8 frames.

Mary cuts the honeycomb into rectangular pieces measuring 11 cm by 7.5 cm

Work out the maximum number of pieces that Mary can get from her **beehive**.

Use 1 inch = 2.5 cm

**[5 marks]**

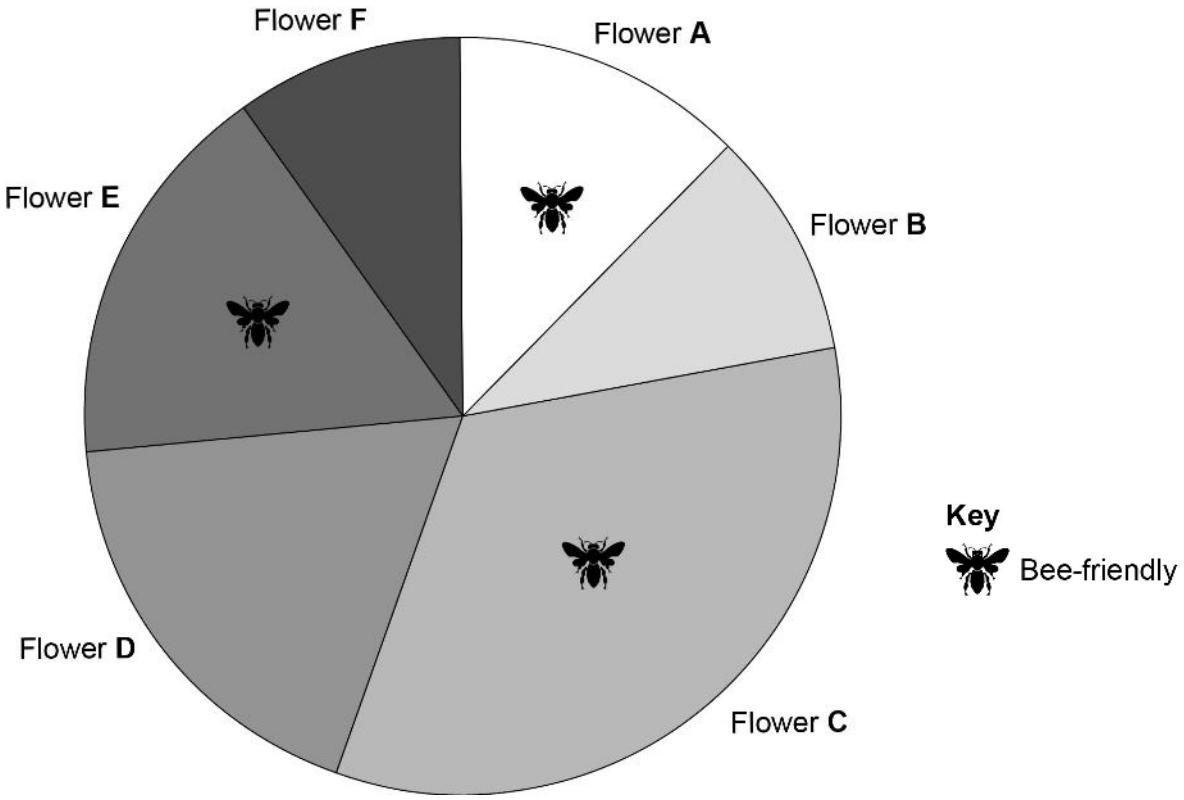
Answer _____

*Do not write outside the box*

**12 (b)** Mary wants to grow some bee-friendly flowers.

She finds information about the different flowers produced from two packets of seeds.

**Packet 1**




Two thirds of the seeds in **Packet 2** produce bee-friendly flowers.

Mary wants to buy the packet producing the greater proportion of bee-friendly flowers.

Which packet should she buy?

You **must** show your working.

**[4 marks]**

*Do not write outside the box*

**12 (c)** Here are the instructions for planting flower seeds.

Use 4 grams of seed per square metre of garden.

Mix the seeds with sand in the ratio

mass of seed : mass of sand = 2 : 5

Mary measures her neighbours' gardens to work out the average-sized garden.

| **Garden ($m^2$)** | **Frequency** | **Midpoint** | |
|---|---|---|---|
| $0 < \text{area} \leqslant 10$ | 2 | | |
| $10 < \text{area} \leqslant 20$ | 8 | | |
| $20 < \text{area} \leqslant 30$ | 12 | | |
| $30 < \text{area} \leqslant 40$ | 3 | | |
| | | | |

Estimate the total mass of seed and sand mix needed to cover an average-sized garden.

**[6 marks]**

Answer _____ grams

**15**

**END OF QUESTIONS**

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**ANSWER IN THE SPACES PROVIDED**

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| Question number | **Additional page, if required.**<br>**Write the question numbers in the left-hand margin.** |
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